Light

GCSE Computer Science - Legal, Ethical and Environmental Issues (Set 2)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset ← Back to Legal Ethical Environmental

Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What does the data minimisation principle of the Data Protection Act 2018 require?

- ○ **A.** Organisations should collect only the personal data that is necessary for the specified purpose
- ○ **B.** Organisations must delete all data immediately after collection
- ○ **C.** Personal data should be stored on the smallest possible hard drive
- ○ **D.** Companies must minimise their number of employees who use computers

Q2 (1 mark)

Under Section 3A of the Computer Misuse Act 1990, what is illegal?

- ○ **A.** Using antivirus software to protect your computer
- ○ **B.** Making, supplying, or obtaining tools intended for use in computer crimes
- ○ **C.** Learning about computer security in school
- ○ **D.** Reporting security vulnerabilities to companies

Q3 (1 mark)

What is Creative Commons licensing?

- ○ **A.** A type of software that creates artwork automatically
- ○ **B.** A government department that registers copyrights
- ○ **C.** A set of copyright licences that allow creators to specify how others may use their work, from full sharing to restricted use
- ○ **D.** A company that sells creative software

Q4 (1 mark)

What is a key requirement of the GNU General Public License (GPL)?

- ○ **A.** If you modify and distribute GPL software, you must also release your modifications under the GPL (copyleft)
- ○ **B.** All GPL software must be sold commercially
- ○ **C.** GPL software can only be used on Linux computers

- ○ **D.** Users must pay royalties to the original developer

Q5 (1 mark)

What is cyberbullying?

- ○ **A.** A type of computer virus that damages files
- ○ **B.** A method of securing computer networks
- ○ **C.** A programming technique for creating websites
- ○ **D.** Using digital technologies like social media, messaging, or gaming platforms to deliberately harass, threaten, or humiliate others

Q6 (1 mark)

What is planned obsolescence in technology?

- ○ **A.** Planning to recycle old devices
- ○ **B.** A method of backing up data before equipment fails
- ○ **C.** Deliberately designing products with limited lifespans so consumers must replace them sooner, generating more sales but increasing waste
- ○ **D.** Planning for software updates

Q7 (1 mark)

What is an online echo chamber?

- ○ **A.** A website for recording and playing back audio
- ○ **B.** An online environment where users primarily encounter information and opinions that reinforce their existing beliefs
- ○ **C.** A type of computer speaker system
- ○ **D.** A secure chatroom for private conversations

Q8 (1 mark)

Why is location tracking by smartphone apps a privacy concern?

- ○ **A.** Location data reveals where you go, when, and how often - this sensitive information can be sold, hacked, or used for surveillance
- ○ **B.** It makes phones heavier
- ○ **C.** GPS technology uses too much storage space
- ○ **D.** Location tracking only works outdoors

Q9 (1 mark)

Under the Data Protection Act 2018, organisations must have a lawful basis to process personal data. Which is a valid lawful basis?

- ○ **A.** The organisation wants to make more profit
- ○ **B.** The data was easy to collect
- ○ **C.** Other companies are doing it

○ **D.** Consent - the individual has given clear permission for their data to be processed for specific purposes

Q10 (1 mark)

Why is digital accessibility an ethical issue in technology?

○ **A.** Accessible websites are more expensive to build

○ **B.** Accessibility only matters for government websites

○ **C.** People with disabilities have the right to access digital services; inaccessible technology excludes them from education, employment, and services

○ **D.** Screen readers are too complicated to support

Q11 (1 mark)

What are conflict minerals in the context of technology manufacturing?

○ **A.** Minerals that react dangerously with each other

○ **B.** Minerals that conflict with other computer components

○ **C.** Artificial minerals created in laboratories

○ **D.** Minerals mined in conditions that fund armed conflict, involve human rights abuses, or cause environmental destruction

Q12 (1 mark)

How does streaming copyrighted content differ legally from downloading it illegally?

- ○ **A.** Streaming from legitimate services like Netflix is legal because the service holds licences; illegal downloading creates permanent copies without licence
- ○ **B.** There is no legal difference between streaming and downloading
- ○ **C.** All streaming is illegal regardless of the source
- ○ **D.** Downloading is always legal if you delete the file afterwards

Q13 (2 marks)

A customer submits a Subject Access Request (SAR) to a company, asking for all personal data held about them.
What must the company do under the Data Protection Act 2018?

- ○ **A.** The company can ignore the request if it is inconvenient
- ○ **B.** The company must provide all personal data held about the individual within one month, free of charge, along with information about how the data is being used
- ○ **C.** The company only needs to provide data if the customer pays a large fee
- ○ **D.** Subject Access Requests only apply to government organisations

Q14 (2 marks)

A group launches a Distributed Denial of Service (DDoS) attack against a business website, making it unavailable to customers for several hours.
Which offence under the Computer Misuse Act 1990 does this constitute?

- ○ **A.** Section 3: Unauthorised acts with intent to impair operation of a computer - flooding a system with traffic to prevent legitimate access impairs its operation
- ○ **B.** No offence - websites going offline is a normal technical issue
- ○ **C.** Only a civil matter for the business to resolve
- ○ **D.** Section 1 only - because no data was accessed

Q15 (2 marks)

A student quotes two paragraphs from a published book in their essay, with proper citation and reference to the original author.
Is this legal under UK copyright law?

- ○ **A.** No - any copying of copyrighted work is always illegal
- ○ **B.** Only if the student pays royalties to the author
- ○ **C.** Yes - this likely falls under 'fair dealing' for research and private study; small portions used for criticism, review, or education with attribution are permitted exceptions to copyright
- ○ **D.** Only if the book is more than 100 years old

Q16 (2 marks)

When installing software, users are asked to accept an End User Licence Agreement (EULA).
What is the purpose and effect of this agreement?

- ○ **A.** EULAs have no legal effect and can be ignored
- ○ **B.** EULAs only apply to free software
- ○ **C.** Accepting a EULA transfers copyright ownership to the user

○ **D.** The EULA is a legal contract defining how the software can be used, what is prohibited, liability limitations, and the rights retained by the software owner

Q17 (2 marks)

The rapid spread of misinformation on social media during elections is considered an ethical concern.
Why is this problematic for democratic society?

○ **A.** False information can manipulate voter opinions, undermine trust in legitimate sources, and distort democratic processes by preventing informed decision-making

○ **B.** Social media platforms earn too much advertising revenue

○ **C.** Politicians should not use social media at all

○ **D.** Misinformation only affects young voters

Q18 (2 marks)

A company moves from local servers to cloud computing, claiming it is more environmentally friendly.
Evaluate this claim.

○ **A.** Cloud computing has no environmental impact because data is stored 'in the cloud'

○ **B.** Cloud computing can be more efficient due to economies of scale and optimised data centres, but the environmental impact still exists and depends on the provider's energy sources and efficiency

○ **C.** Local servers are always more environmentally friendly than cloud

- ○ **D.** Cloud computing uses no electricity

Q19 (2 marks)

An individual requests that a search engine remove links to old news articles about them under the 'right to be forgotten'.
Under what circumstances might this request be granted?

- ○ **A.** All such requests must always be granted immediately
- ○ **B.** The right to be forgotten does not exist in any law
- ○ **C.** The request may be granted if the information is outdated, irrelevant, or causing disproportionate harm, balanced against public interest in the information remaining accessible
- ○ **D.** Only celebrities can request removal of search results

Q20 (2 marks)

Online learning platforms have made educational content accessible globally.
What cultural impact does this have?

- ○ **A.** Online education has no cultural impact
- ○ **B.** Only universities can provide online education
- ○ **C.** Online learning is identical to classroom learning
- ○ **D.** It democratises access to knowledge for those with internet access, but may also spread dominant cultural perspectives while marginalising local educational traditions and languages

Q21 (2 marks)

A software developer discovers their company's product has a security flaw that could expose customer data.
What ethical responsibilities do they have?

- ○ **A.** They have an ethical duty to report the vulnerability internally, advocate for fixing it promptly, and consider whistleblowing if the company ignores serious risks to users
- ○ **B.** No ethical responsibility - security is not a developer's concern
- ○ **C.** They should sell the vulnerability information to hackers
- ○ **D.** They should hide the flaw to protect company profits

Q22 (2 marks)

What is the difference between freeware and open source software?

- ○ **A.** There is no difference - both terms mean the same thing
- ○ **B.** Freeware is free to use but source code is not available; open source provides free access to source code for viewing, modification, and distribution
- ○ **C.** Freeware is always better quality than open source
- ○ **D.** Open source software always costs money

Q23 (2 marks)

Why do data protection laws include special protections for children's personal data?

- ○ **A.** Children do not use technology, so no protection is needed

- ○ **B.** Children's data is less valuable to companies
- ○ **C.** Children may not fully understand privacy implications; their data requires extra protection because they are more vulnerable and less able to give meaningful consent
- ○ **D.** These protections only apply to children under 5

Q24 (3 marks)

A disgruntled employee installs a 'logic bomb' in company software - code that will delete important files six months after they leave. The code is discovered before it activates.
Analyse which Computer Misuse Act offences apply.

- ○ **A.** No offence until the code actually activates and causes damage
- ○ **B.** Section 3 (unauthorised acts with intent to impair) applies even though it has not activated - the intent to impair future operation is established; Section 3A may also apply for creating the malicious code; the employee's authorised access does not authorise this specific malicious action
- ○ **C.** This is only an employment matter, not a criminal offence
- ○ **D.** Logic bombs are legal if created by employees

Q25 (3 marks)

Autonomous vehicles must be programmed to make decisions in unavoidable accident situations.
Analyse the ethical challenges of programming such decisions.

- ○ **A.** Autonomous vehicles never have accidents so this is not a real issue
- ○ **B.** The vehicle should always protect the driver regardless of other

consequences

○ **C.** Only humans can drive vehicles so this issue does not exist

○ **D.** Programming responses to unavoidable accidents raises difficult ethical questions: should the car prioritise passengers or pedestrians? How should it weigh different lives? Who is morally and legally responsible for these programmed choices? Can machines make ethical decisions?

Q26 (3 marks)

Manufacturers are increasingly making devices difficult to repair: using proprietary screws, gluing components together, and not providing spare parts.
Analyse the environmental and consumer rights implications.

○ **A.** Difficult repairs have no negative consequences

○ **B.** Repair restrictions create e-waste by forcing replacement of fixable devices; deny consumers the right to repair their property; benefit manufacturers through forced upgrades; the 'right to repair' movement advocates for legislation requiring accessible repairs, spare parts availability, and repair documentation

○ **C.** Consumers have no interest in repairing devices

○ **D.** Only manufacturers should repair devices for safety reasons

Q27 (3 marks)

A UK company uses a cloud service provider based in the United States to store customer data.
Analyse the Data Protection Act compliance considerations for this international data transfer.

○ **A.** Data protection laws only apply within the UK, so international transfers have no restrictions

○ **B.** Cloud storage automatically complies with all data protection laws

○ **C.** Only physical paper records are covered by data protection law

○ **D.** International transfers require safeguards: adequate protection in the destination country or contractual clauses ensuring equivalent protection; the company remains responsible for customer data regardless of where it is processed; different jurisdictions may have conflicting legal access requirements

Q28 (3 marks)

Social media allows people to create and control their online identities differently from their offline selves.
Analyse the cultural and psychological implications.

○ **A.** Online and offline identities are always identical

○ **B.** Only young people use social media

○ **C.** People can experiment with identity expression and find communities; however, curated presentations may create unrealistic comparisons causing anxiety; anonymous identities may enable harmful behaviour; boundaries between authentic and performed identity can blur

○ **D.** Online identity has no effect on real life

Q29 (3 marks)

AI systems are trained on vast datasets of text and images scraped from the internet, then generate new content.
Analyse the copyright implications.

○ **A.** AI-generated content has no copyright implications whatsoever

○ **B.** Complex unresolved questions arise: training on copyrighted works without permission may infringe rights; generated outputs may be derivative works; ownership of AI-generated content is unclear; existing copyright law was not designed for these scenarios; legal frameworks are still developing

○ **C.** AI cannot generate creative content

○ **D.** All AI-generated content is automatically copyrighted to the AI company

Q30 (3 marks)

Smart home devices (voice assistants, smart TVs, connected cameras) continuously collect data about household activities.
Analyse the privacy implications of living in a 'smart home'.

○ **A.** Smart devices only collect data when explicitly asked to do something

○ **B.** Smart home data is always encrypted and completely secure

○ **C.** Privacy laws prevent any data collection by smart devices

○ **D.** Always-on devices create detailed profiles of private home life: conversations, viewing habits, daily routines, presence patterns; this data may be accessed by companies, law enforcement, or hackers; intimate domestic spaces become sites of commercial surveillance; security vulnerabilities can enable intrusion

Total: **55** marks • Time: **30 min**

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